

✓ Lab 1 — Exercise: Notebooks & Markdown

ITCS 3162 — Introduction to Data Mining

Name: *your name here* **Date:** *today's date*

Complete the exercises below. For each task:

- **Code tasks** → fill in the code cell
- **Written answers** → edit the markdown cell directly (look for `YOUR ANSWER:` prompts)

When you're done, run **Kernel → Restart & Run All** to make sure the whole notebook executes top-to-bottom without errors, then download the `.ipynb` and submit via Canvas.

✓ Exercise 1 — Running cells (5 pts)

Run the cell below. Then **modify** it so that it prints your full name and your major.

```
# TODO: change the strings to your name and major
name = "Alex"
major = "Data Science"
print(f"Hi, I'm {name} and I'm studying {major}.")
```

```
Hi, I'm Alex and I'm studying Data Science.
```

✓ Exercise 2 — Variable persistence (10 pts)

Write code in the next cell that creates two variables, `a` and `b`, with any numeric values you like. Then in the cell after that, print their sum, difference, and product on three separate lines.

```
# TODO: define a and b
a = 6
b = 7
```

```
# TODO: print a+b, a-b, a*b on separate lines
print (a+b)
print (a-b)
print (a*b)
```

```
13
-1
42
```

Exercise 3 — Markdown formatting (15 pts)

Double-click this cell and replace the placeholder content below with:

1. A level-2 heading with the text "About me"
2. A bulleted list with three items: your favorite class, a hobby, and a goal for this course
3. One sentence containing at least one **bold** and one *italic* word
4. An inline code reference to `pandas`
5. A link to a website you find useful

About me

- My favorite class is any STAT class
- I enjoy video games and weight training
- My goal for this course is to earn an A and to use this course to prevent skill atrophy during my summer internship

Video of *Will Smith* eating spaghetti.

<https://app.datacamp.com/>

Exercise 4 — Math notation (10 pts)

Replace this cell's content with the LaTeX rendering of the **slope-intercept form** of a line, displayed (not inline). Then below it, write one sentence using inline math to explain what m represents.

YOUR ANSWER:

$$y = mx + b$$

the m is the slope of the line

Exercise 5 — A small table (10 pts)

In the cell below, create a markdown table listing **three data mining tasks** we'll cover this semester. Columns: `Task`, `Type` (supervised / unsupervised / association), and `Example use case`.

YOUR ANSWER:

Task	Type	Example use case
Classification	Supervised	Predicting whether an email is spam

Task	Type	Example use case
Clustering	Unsupervised	Grouping customers by shopping behavior
Association Rules	Association	Recommending products frequently bought together

TT **B** *I* <> ↺ 🖼️ “ ☰ ☷ — Ψ 😊 ☰ Close

```
## Exercise 6 – Reflection (10 pts)

In 2–4 sentences, describe one thing you found confusing about notebooks and one thing you find useful compared to writing a regular Python script.

YOUR ANSWER:

I found making tables confusing. I like how I can use multiple cells while I code. I feel like that it will help me sort my code better.
```

Exercise 6 — Reflection (10 pts)

In 2–4 sentences, describe one thing you found confusing about notebooks and one thing you find useful compared to writing a regular Python script.

YOUR ANSWER:

I found making tables confusing. I like how I can use multiple cells while I code. I feel like that it will help me sort my code better.

Submission checklist

- ☐ Replaced your name and date at the top
- ☐ All code cells run without errors
- ☐ All `YOUR ANSWER` prompts have been replaced
- ☐ Ran **Restart & Run All** before exporting
- ☐ Downloaded as `.ipynb` and uploaded to Canvas

